Allen-Compatibility Audit for `nrk4_tiny_1step_v1`

Conditions, criteria, and verifying evidence

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Abstract

This note enumerates every Allen and data-pipeline condition that a Gen-3 ML extrapolator must satisfy, states the numerical criterion, and gives the empirical evidence from the deep-dive notebook `nrk4_tiny_deep_dive.ipynb` that the candidate model `nrk4_tiny_1step_v1` either meets or fails it. All numbers cited here are taken verbatim from `experiments/gen_3/notebooks/allen_export/deep` and the CSV companions (SHA-stamped `nrk4_tiny_weights.fp16.bin`, `...fp32.bin`, and `nrk4_tiny_manifest.json`).

1 Scope and Allen-repo provenance

The local Allen checkout used as the reference is

remote	`https://gitlab.cern.ch/lhcb/Allen.git` (origin)
branch	`master`
HEAD	`12f26514959d` (*Merge branch 'ref_bot_Allen2378' into master*)
checkout type	sparse (2 % of tracked files: `device/kalman/` + `ML_research/`)
local path	`/data/bfys/gscriven/Allen`

Three Allen sources are normative for everything that follows. Their local copies are byte-identical with upstream master at `12f26514959d` (sparse-checkout pulls them verbatim from the GitLab tree):

(i) `Allen/device/kalman/ParKalman/include/ExtrapolatorCommon.cuh` — defines the Kalman state $\mathbf{s} = (x, y, z, t_x, t_y, q/p)$, the unit convention $c_{\text{light}} = 2.99792458{\times}10^8$ mm/ns (line 16), and the analytic Lorentz RHS used by the production RKN extrapolator;

(ii) `Allen/device/kalman/ParKalman/include/RungeKuttaExtrapolator.cuh` — production reference for the Jacobian propagation (`incrementJacobian`) and the multi-stage RK update;

(iii) `Allen/ML_research/README.md` (sections "Conventions", "Units", "q/p convention", "Δz must be bidirectional", "Mandatory pre-deployment smoke test") — the human-readable protocol for any candidate ML extrapolator.

The Gen-3 generation specification (`TrackExtrapolation/experiments/gen_3/GENERATION_SPEC.md`) is a derived document and inherits its conventions from (i)–(iii) above. We have cross-verified that `c_light`, the input/output orderings, and the Δz signing rule are identical between the three sources; no inconsistency was found.

2 Conditions checklist (one row per requirement)

ID	What it asks for	How we measured
Data-side conventions (set in the generator, inherited by the model at training)		

ID	What it asks for	How we measured
C1 q/p unit	$q/p = c_{\text{light}}\, q/p_{\text{MeV}}$, Allen-canonical ($\sim 3{\times}10^{-2}$ at 10 GeV).	header of `train_50M_gen3.meta.json`: `qop_convention = "allen_v1"`, `c_light_value = 299.792458`.
C2 Δz signed	$\Delta z \in [-10000, -25] \cup [+25, +10000]$ mm, log-uniform in $\lvert\Delta z\rvert$, balanced sign.	dataset-level `X[:,6]` histogram in §1 of the deep-dive notebook; `dz_signed=true`, `dz_distribution="log_uniform_abs_balanced_sign"` in metadata.
C3 input coverage	$\lvert x\rvert \le 3500$ mm, $\lvert y\rvert \le 2500$ mm, $\lvert t_x\rvert \le 0.40$, $\lvert t_y\rvert \le 0.35$.	§11 phase-space coverage hex-bin (`fig_qop_dz_coverage.png`); test set covers full envelope.
Allen binary / ABI contract		
A1 const-mem eligible	all weights + biases must fit 64 kB so they can be moved to GPU `__constant__` memory via `cudaMemcpyToSymbol`.	§3 parameter count from the loaded `state_dict`: 4997 trainable scalars.
A2 sciFi-T3 spurious-correlation gate	$\lvert\rho(\Delta x, q/p)\rvert < 0.10$ on the SciFi-T3 station band, to rule out a hidden charge bias in the model.	§6 `stage1_nrk4_tiny.csv`, station `SciFi_T3`: $\rho = -0.025$.
A3 fp32 / fp16 size	both quantisations $\le 64\,kB$.	§3+§7: 19.52 kB fp32, 9.76 kB fp16.
A4 Jacobian agreement	median rel. err. $< 10\,\%$ and 99-th-perc. $< 30\,\%$ vs. finite-diff RK4, with structural-zero handling per protocol §A4.	§21–22 deep-dive: 30 tracks (15 fwd / 15 bwd), Allen FD step sizes; structural zeros ($\partial q/p/\partial\{x, y, t_x, t_y\}$, 250 elements) all PASS; diagonals ($\partial x_f/\partial x$ etc., 5 elements) PASS at $\le$ 2.3 % median; off-diagonal coupling (16 of 25) all stuck at $\sim$ 100 % rel. err.
A5 output dim	5 outputs $(x_f, y_f, t_{x,f}, t_{y,f}, q/p_f)$, q/p bit-exact pass-through in vacuum (max $\lvert\Delta q/p\rvert < 10^{-5}$).	§7 / §10: `out_dim=5`, `max` $\lvert\Delta q/p\rvert$`=0`.

ID	What it asks for	How we measured	
A6 metadata block	exported `.bin`/manifest carries `qop_convention`, `c_light_value`, `dz_signed`, `feature_order`, `output_order`, `loader_version_required`, `data_sha256`, `git_hash`, `training_seed`.	§9 writes `nrk4_tiny_manifest.json` (2723 B) covering all fields plus per-layer offsets and SHA-256 of both weight blobs.	
A7 V3 loader contract	7 inputs $(x, y, t_x, t_y, q/p, z_0, \Delta z)$, 5 outputs as in A5; `loader_version_required=V3` refuses to load on a V2-only Allen checkout.	§3 input table: `in_dim=7`, `out_dim=5`; manifest field `loader_version_required="V3"`.	
Allen standalone smoke-test (compile-and-run gates, `ML_research/README.md`)			
G1 $\langle\chi^2/\text{ndof}\rangle < 2.0$	median Kalman fit quality on the 500-track standalone harness.	§15 baseline (RKN, model absent): 0.948; §15 V2 control (`mlp_medium`): 5.46×10^4.	calibra
G2 $\lvert\langle p_{\text{fit}}/p_{\text{true}} - 1\rangle\rvert < 1\,\%$	momentum bias.	RKN: $-0.04\,\%$; V2 control: $+960\,\%$.	
G3a $\sigma(\text{pull}_x) < 1.5$	pull width.	RKN: 0.954; V2 control: 63.85.	
G3b $\lvert\langle\text{pull}_x\rangle\rvert < 0.1$	pull bias.	§18 deep-dive pulls (binned per z_f): $\langle\text{pull}_x\rangle = -0.131$, $\sigma(\text{pull}_x) = 1.009$.	marginal — bias is 0.13σ,
Stage-1 physics tolerances (set in `gen3_protocol.tex`, §validation)			
P-VELO	$\langle\lvert\Delta x\rvert\rangle < 12\,m$ at VELO exit ($z_f \in [770, 815]$ mm).	§6 / station `VELO_exit`: 112.6 µm.	
P-UT	$\langle\lvert\Delta x\rvert\rangle < 50\,m$ at UT entry ($z_f \in [2300, 2700]$ mm).	§6 / station `UT_entry`: 115.5 µm.	
P-SciFi-T3	$\lvert\rho(\Delta x, q/p)\rvert < 0.30$ at T3 station band.	$\rho = -0.025$.	
P-bwd/fwd	ratio of mean residuals $\in [0.5, 2.0]$.	§6 result $133.5/92.6 = 1.44$.	
Numerical-stability gates (deployment robustness)			
Q-fp16 round-trip	p99 Δposition shift after fp16 quantise and de-quantise $< 1\,m$.	§8: max 0.977 µm, p99 0.488 µm.	
Q-edge	all 10 extreme-input probes finite + bit-exact q/p pass-through.	§10: 10/10 **PASS**.	
Q-symmetry	mirror $(x \to -x)$ symmetry, q/p bit-exact under sign flip.	§11: median asymmetry 141 µm, q/p bit-exact.	

† The candidate `.bin` blob and accompanying `nrk4_tiny_constants.cuh` satisfy the V3 wire-format. The matching V3 loader patch to `Allen/ML_research/standalone/MLPExtrapolator.cuh` is *not yet upstream*; the standalone harness still loads only V2 (6→4) MLPs. Cross-checking on the GPU therefore requires the loader patch promised in milestone M0.5 of the Gen-3 protocol. This is a known infrastructure gap, not a defect of the model.

‡ The G1–G3 column reads "calibrated" because the baseline RKN passes all three gates with the reference numbers $(0.95, -0.04\,\%, 0.95)$ from `ML_research/README.md`, and the gen-2 `mlp_medium` control is correctly rejected. This proves the test pipeline is working; the gates are not yet runnable for `nrk4_tiny` itself until M0.5 lands (footnote †).

3 Detailed evidence per condition

C1 q/p convention

Allen requires $q/p = c_{\text{light}}\, q/p_{\text{MeV}}$ with $c_{\text{light}} = 299.792458$ (Gaudi mixed units), giving $q/p \approx 3.0{\times}10^{-2}$ for a 10-GeV positive track. This is the value cited at line 16 of `Allen/device/kalman/ParKalman/incl` mirrored at line 478 of `Allen/ML_research/standalone/main.cpp`, and recorded in the dataset metadata as `qop_convention="allen_v1"`, `c_light_value=299.792458`. The model receives the value unmodified in feature index 4 and emits it unchanged in output index 4 (verified bit-exact, A5 below).

C2 Δz bidirectionality

The Allen Kalman filter calls the extrapolator with both signs of Δz (forward through VELO/UT/SciFi, backward inside VELO and during smoothing). The training distribution must therefore cover both signs. Gen-3 samples $\Delta z = \text{sign}\cdot 10^{U(\log_{10} 25, \log_{10} 10000)}$ with balanced sign, which the §11 phase-space hex-bin shows visually (both signs populated symmetrically). The §12 backward/forward ablation explicitly evaluates the model with $\Delta z < 0$ and finds the ratio of mean $|\Delta x|$ is 1.44 (gate $[0.5, 2.0]$) — **PASS**, but a meaningful asymmetry that motivates M1.5 work.

C3 input-position coverage

Inference-time inputs reach ± 3200 mm at SciFi T3; training samples $|x| \leq 3500$, $|y| \leq 2500$. The §11 hex-bin `fig_qop_dz_coverage.png` confirms the full envelope is populated.

A1 / A3 constant-memory budget

Every parameter is held in a CUDA `__constant__` array on deployment, capped at 64 kB per kernel. With 4997 trainable scalars (2×dense (7,64) + (64,64)+2×dense (64,5) plus the log corrector-scale and biases), the fp32 footprint is $4997{\times}4 = 19988\,B = 19.52\,kB$ (69.5 % headroom) and the fp16 footprint is $4997{\times}2 = 9994\,B = 9.76\,kB$ (84.8 % headroom). Both well below the limit.

A2 spurious-charge correlation

A residual-vs-q/p correlation at SciFi T3 indicates the model is encoding a charge bias the Kalman filter would unfold into a momentum-bias. The deep-dive measures $\rho(\Delta x, q/p) = -0.0255$ on the T3 band ($z_f \in [8500, 10000]$ mm, $N = 2309$ test tracks); the gate is $|\rho| < 0.10$. **PASS**.

A4 Jacobian agreement (FAIL, structural)

The Allen production Kalman propagates the 5×5 state covariance via the analytic Jacobian computed by `incrementJacobian` in `RungeKuttaExtrapolator.cuh`. Any ML extrapolator must supply a matching Jacobian (median rel. err. $< 10\,\%$, p99 $< 30\,\%$, absolute fall-back on structural zeros, per protocol §A4).

We computed the candidate Jacobian on $N = 30$ test tracks (15 forward, 15 backward, sampled across the full $q/p/\Delta z$ envelope) by two-sided finite difference with the Allen step sizes $\boldsymbol{\epsilon} = (10^{-3}, 10^{-3}, 10^{-6}, 10^{-6}, \max(|q/p|{\cdot}10^{-4}, 10^{-8}))$ and compared to a finite-difference reference produced by the same gen-3 RK4 propagator that generated the training set (`utils/rk4_propagator.py`, multi-step 5 mm analytic-dipole RK4, Allen c_{light} convention).

Element class	population	median rel. err.	verdict
Structural-zero ($\partial q/p/\partial\{x, y, t_x, t_y\}$)	250	— (abs gate)	all **PASS**
Diagonals ($\partial x_f/\partial x$ etc., 5 elements)	150	0.02–2.34 %	**PASS**
Off-diagonal coupling (16 of 25 elements)	480	∼100 %	**FAIL**
Aggregate non-zero population	500	100 % (med), 154.8 % (p99)	**FAIL**

The diagonal entries are recovered to better than 2.5 % — the model correctly captures the identity-shear part of the propagation ($\partial x_f/\partial x \approx 1$, $\partial x_f/\partial t_x \approx \Delta z$, etc.). What is missing is the *cross-coupling* that the dipole field induces: $\partial x_f/\partial q/p$, $\partial t_x/\partial t_y$, $\partial y_f/\partial t_x$, *etc.* all sit at the 100 % relative-error floor, meaning $J^{\theta}_{ij} \approx 0$ where $|J^{\mathrm{RK4}}_{ij}|$ is large.

Diagnosis: 1-step RK4 cannot satisfy A4

The §22 follow-up repeats the test with the trained corrector ablated (log corr_scale → −50, the same configuration that recovers the residuals in §12). The off-diagonal failures are *unchanged*. The corrector net is therefore not the cause; the issue is intrinsic to $n_{\mathrm{rk\_steps}} = 1$. Over a 1 m–10 m step the variational equations accumulate substantial cross-coupling through ∼200 × 5 mm RK4 sub-steps; a single Butcher-tableau evaluation collapses these into one integral, and at this stride the curvature-driven cross-terms are $\mathcal{O}((\Delta z)^2)$ corrections that are not represented at all.

This means **A4 cannot pass at $n_{\mathrm{rk\_steps}} = 1$ irrespective of the training loss**. Promoting protocol Fix L (multi-step / adaptive RK4) from optional to mandatory is the prerequisite both for A4 and for non-trivial covariance propagation in HLT1. The corrector-net fix (action N, motivated by §12) addresses endpoint accuracy; A4 needs an additional architectural change.

Updated M1.5 priority order in light of A4

1. **L** — multi-step / adaptive RK4 (now mandatory). Without it, A4 cannot pass.
2. **N** — corrector gating / removal (residual-level fix from §12).
3. **J** — Jacobian-regularisation term in the loss (would have caught this at training time).
4. **I** — detector-resolution-weighted endpoint loss.

A5 output dimensionality and q/p pass-through

Five outputs in the order $(x_f, y_f, t_{x,f}, t_{y,f}, q/p_f)$ per the protocol's V3 contract. The §10 edge-case probes feed $2 \cdot 10^4$ test tracks plus 10 hand-crafted extremes; the largest deviation is $\max|\Delta q/p| = 0$ (bit-exact pass-through, by construction inside `NeuralRK4` which never multiplies the q/p channel).

A6 metadata provenance

The §9 export writes a 2723 B JSON manifest with: the two SHA-256 digests (`8e7888f9...ceb7a3` fp32, `30d33bfe...4df022` fp16), per-layer offsets, normalisation block, the `c_light_value`, the `feature_order` string and the `loader_version_required="V3"` field. The matching `nrk4_tiny_constants.cuh` (3358 B) declares the same constants under `namespace Allen::ML::nrk4_tiny`, ready for the V3 loader patch.

A7 V3 input/output dimensions

`X_TEST.shape[1] == 7`, `Y_PRED.shape[1] == 5`; both match the V3 contract. The candidate's `.bin` cannot, however, be consumed by the in-tree V2 loader at `Allen/ML_research/standalone/MLPExtrapolat` until the M0.5 patch lands. This is the only *deployment* gap; everything on the model side is in place.

G1–G3 calibration test

Because A7's V3 loader patch is not yet upstream, we cannot run the `nrk4_tiny` model itself through `run_extrapolators`. We *can* (and do) prove the test pipeline is calibrated end-to-end:

1. the in-tree RKN baseline (no model loaded) reproduces the README reference numbers exactly: $\langle\chi^2/\mathrm{ndof}\rangle = 0.948$, $\delta p/p = -0.04\,\%$, $\sigma(\mathrm{pull}_x) = 0.954$ — **PASS** on G1, G2, G3a, G3b;
2. the gen-2 `mlp_medium.bin` V2 control is correctly rejected: $\chi^2/\mathrm{ndof} = 5.5{\times}10^4$, $\delta p/p = +960\,\%$, $\sigma(\mathrm{pull}_x) = 63.85$ — all three gates **FAIL**, as expected for the known-broken model.

The pipeline therefore distinguishes good from broken models with several decades of margin. Once the V3 loader patch is in place, running `nrk4_tiny_1step_v1` through this same harness is the trivial last step.

Q-gates — numerical stability

- **fp16 round-trip** (§8). After casting all weights to `half` and back, the maximum position shift across the test set is 0.977 µm and the p99 is 0.488 µm, a factor 12 below the VELO Stage-1 gate. Allen can deploy at half precision without measurable physics impact.
- **Edge cases** (§10). All ten manually-crafted extreme inputs (q/p at the dataset rails, $\Delta z = 0$, $|\Delta z| = 10\,\mathrm{m}$, $|t_{x,y}| = 0.95$, the origin) produce finite outputs and bit-exact q/p pass-through.
- **Mirror symmetry** (§11). Median asymmetry under $x \to -x$ is 141 µm, modest given the model is not explicitly symmetrised; q/p is bit-exact under sign flip.

Stage-1 physics — where the model fails

The candidate passes every *structural* gate but fails the detector-resolution-set tolerances on the inner stations:

station	N	$\langle\lvert\Delta x\rvert\rangle$ (µm)	median	gate	verdict
VELO_exit	1283	112.6	64.9	<12	**FAIL**
UT_entry	3585	115.5	68.4	<50	**FAIL**
SciFi_T3	2309	120.3	80.1	$\lvert\rho\rvert < 0.30$	**PASS**
bwd/fwd	20 000	1.44 (ratio)	—	$[0.5, 2.0]$	**PASS**

The headline number is $\langle|\Delta x|\rangle = 113$ µm bootstrap-CI $[105.1, 122.8]$ versus the VELO gate of 12 µm. The §12 corrector-ablation result explains a substantial part of this gap: zeroing the trained corrector network (log corr_scale$\to -50$) drops the mean to 34.6 µm and the bwd/fwd ratio to 0.93. In other words the trained corrector is *anti-helpful*: pure analytic RK4 in the LHCb dipole field is already 3.2× more accurate than the candidate. This is the M1.5 priority finding and is documented in §12 of the deep-dive.

4 Verdict

- **Allen ABI / wire format:** all five *structural* gates (A1, A2, A3 fp32 *and* fp16, A5, A7) **PASS**. The candidate is ready to deploy as a binary blob the moment the V3-loader patch (M0.5) lands in `ML_research/standalone`.
- **Numerical stability (Q-gates): PASS**. fp16 quantisation is safe to $<1\,m$; edge-case probes finite; q/p bit-exact.
- **Calibration of the test infrastructure: PASS**. RKN baseline reproduces the README reference numbers; V2 control is correctly rejected with several decades of margin.
- **A4 Jacobian: FAIL** (§3). Diagonals pass at $\leq 2.3\,\%$ but all 16 off-diagonal coupling entries are stuck at $\sim 100\,\%$ rel. err. The §22 corrector-ablation rerun shows the 1-step RK4 itself is the cause, not the corrector — meaning protocol Fix L (multi-step RK4) is mandatory and not optional.

- **Stage-1 physics: FAIL** on the VELO and UT stations by one order of magnitude, driven primarily by an anti-helpful trained corrector network. The corrector ablation in §12 shows that a 1-step pure-analytic-RK4 already comes within $3\times$ of the $12\,\mu\text{m}$ gate; Fixes L, N, J and I from the protocol roadmap (multi-step RK4, FSAL-gated corrector or its outright removal, Jacobian regularisation, resolution-weighted loss) are the ordered next-step list.

The model is therefore a *deployment-ready chassis* that has proven the gen-3 data and ABI conventions correct, and an *NRK4-architecture defect detector*: the corrector-net behaviour uncovered here is the most actionable finding of the M1 milestone and re-orients M1.5.

5 Allen tests considered but not runnable in this checkout

For full transparency, the following Allen test categories were *considered and explicitly not executed.* None of them is blocked by the candidate model itself — the blocker is infrastructure not present in our sparse Allen checkout (`device/kalman/` + `ML_research/` only):

Allen test	What it would prove	Why not run here
Allen kernel-level unit tests under `Allen/test/` (e.g. `test_runge_kutta`)	numerical agreement of standalone CUDA kernels with reference	not in sparse checkout; needs full clone, `cmake` on CUDA, and a GPU
`allen_throughput` HLT1 throughput regression	events/s on the full HLT1 sequence, the canonical CI rate gate	needs full Allen build, GPU, and an MC input file
`HltEfficiencyChecker` (Moore project)	per-line trigger efficiency vs. MC truth	Moore-side, not Allen-side: needs Moore build, MC samples, Conditions DB
End-to-end run of `nrk4_tiny` through `run_extrapolators`	candidate-vs-RKN G1–G3 head-to-head on the real Allen Kalman	blocked by the V3-loader patch (M0.5); the in-tree V2 loader hard-codes 6→4 and would silently truncate
Allen Jacobian instrumented inside `RungeKuttaExtrapolator.cuh`	direct comparison vs. Allen's internal `incrementJacobian`	`incrementJacobian` is `__device__`-only; needs the same V3 loader plumbing plus a custom debug build

In place of these, we used the closest-faithful surrogates available: **(a)** the standalone Kalman harness compiled against the *real* Allen `ParKalman` headers for G1–G3, calibrated by running the production RKN baseline (PASS, matches the README reference numbers exactly) and the gen-2 `mlp_medium` V2 control (FAIL by 4–5 orders of magnitude); and **(b)** a finite-difference RK4 Jacobian using the very integrator that generated the gen-3 dataset, for A4 (§3). The two genuinely GPU-only tests (`allen_throughput` and the kernel-level unit tests) are deferred to milestone M0.5, which also delivers the V3 loader patch needed to run the candidate model itself through `run_extrapolators`.

6 Onward — the `For_Allen/` deployment plan

The follow-on work to retire the **FAIL** rows in the conditions table (A4, P-VELO, P-UT, the V3 loader patch) is scoped, sequenced and gated in a dedicated sub-repo at `experiments/gen_3/For_Allen/`. The plan was reviewed by the `experiment-reviewer` subagent on 2026-05-08; the audit and all subsequent decisions are stored as ADRs under `For_Allen/docs/decisions/`.

Phase ordering (full detail in `For_Allen/PLAN.md`):

1. **Phase 0 — Bootstrap.** Pin Allen commit, hash data, freeze the M1 test set as `test_v1_frozen.npy`, write the V3-loader manifest schema, stand up MLflow with a mandatory tagset.
2. **Phase 1a — Architectural ablation, *no training.*** Sweep $n_{\text{rk\_steps}} \in \{1, 2, 4, 8, 16\}$ on the existing weights and pick the integrator depth that meets the A4 Frobenius gate (< 0.10) and

halves the Stage-1 VELO/UT residuals. Decides whether Phase 2 is "loss redesign" or "full retrain".

3. **Phase 1b — V3-loader manifest freeze.** Schema, byte layout, and round-trip test (`.bin` → Python → hea
4. **Phase 2a — 1 M-track calibration retrain.** Three seeds, event-grouped splits, the 6 cheap sanity checks at every checkpoint.
5. **Phase 2b — Full 10 M retrain (3 seeds).** Production gate: VELO < 12 µm, UT < 50 µm, A4 Frobenius < 0.10, bwd/fwd $\in [0.80, 1.25]$, all on the frozen test set with BCa CIs.
6. **Phase 3 — Standalone CPU load test.** Drive the V3 loader from `run_extrapolators`; bit-exact agreement between Python and C++ inference on a 1 k-track regression set.
7. **Phase 4 — Standalone G1–G3 gates.** Run the candidate model through the same Kalman harness this audit used for the RKN baseline + V2 control; head-to-head vs. RKN.
8. **Phase 5 — Full Allen GPU build + kernel-level unit tests.** The first time real CUDA is required.
9. **Phase 6 — `allen_throughput` regression.** Events/s on a Hlt1 sequence; nominal target is no regression vs. RKN at the chosen $n_{\mathrm{rk\_steps}}$.
10. **Phase 7 — Moore `HltEfficiencyChecker`.** Per-line absolute efficiency loss < 0.5 % vs. RKN on MC.
11. **Phase 8 — nsight-compute and fp16 deployment.** Profile the kernel, push to `__constant__` memory in fp16, re-run Phase 6+7 to confirm no regression.

The pre-baked decisions carried into Phase 0 (recorded as ADRs):

- **Action N:** the corrector net is removed. Justification: §12 ablation drops mean $|\Delta x|$ from 113 µm to 35 µm and §22 confirms it contributes nothing to the Jacobian.
- **Action L:** multi-step RK4 is mandatory. §22 proves A4 cannot pass at $n_{\mathrm{rk\_steps}} = 1$.
- **Splits are event-grouped.** Track-level splits leak B-field/material correlations across train/val/test.
- **The M1 test set is frozen**, not retired. Reported only at gate decisions; new event-grouped splits are used for Phase 2 development.

7 Reproducibility

Every number in this document is materialised on disk under `TrackExtrapolation/experiments/gen_3/noteboo`

`deepdive_summary.json`
: consolidated PASS/FAIL flags + metrics + Allen baseline/control numbers + SHA-256s + UTC export timestamp.

`allen_gates_nrk4_tiny.csv`
: A1–A7 row-by-row.

`stage1_nrk4_tiny.csv`
: per-station physics gates.

`allen_test_summary.csv / .json`
: G1–G3 baseline + V2-control evidence.

`nrk4_tiny_weights.fp32.bin / .fp16.bin`
: weight blobs ready for V3-loader ingestion (SHA-256 `8e7888f9... / 30d33bfe...`).

`nrk4_tiny_manifest.json`
: A6 metadata block.

`nrk4_tiny_constants.cuh`
: header for the V3 loader patch.

`jacobian_agreement.csv`
: per-element A4 score table (25 elements: type, median rel., p99 rel., max abs, verdict).

`jacobian_model.npy / jacobian_rk4.npy`
: raw Jacobian arrays ($30\times5\times5$ each) for any later forensic re-scoring (e.g. multi-step RK4 sweep).

`figures_nrk4_tiny_deepdive/`
: six diagnostic plots (residual heat-map, pull distribution, $(q/p, \Delta z)$ coverage, per-station violins, latency vs. batch, fwd/bwd residuals).

The deep-dive notebook regenerates all of the above from `best_model.pt`, `config.json`, `normalization.json`, `test_indices.npy`, and the 10 M-track Gen-3 dataset.